Classical Field Theory — Exercise Sheet 2

Noether currents and their limits

Solutions

Use Lecture 2's signs $j^\mu = p_a^\mu Q^a - B^\mu$ and $\partial_\mu j^\mu = -E_a Q^a$. All fields and coefficients below are smooth. The unindexed $Q(t)$ denotes an integrated charge, and Q^a denotes a characteristic. The last problem is a different field theory from Klein–Gordon.

Exercise 2.1 (An improvement can change a boundary charge). Let j^μ be a current in $1+d$ dimensions and let $K^{\mu\nu} = -K^{\nu\mu}$ be smooth. Define $\widetilde{j}^\mu = j^\mu + \partial_\nu K^{\mu\nu}$. An added current with identically zero divergence is called an *identically conserved improvement* in this problem.

(a) Prove that the two currents have the same divergence for every field. Show that their charges agree on the spatial torus when K is periodic. On $\mathbb{R}^d$, assume the original slice charge exists and show equality when $K^{0i}(t,\cdot)$ is compactly supported.

(b) In $1+1$ dimensions take $K^{01} = t/(1+x^2)$ and $K^{10} = -K^{01}$. Compute the added density and flux. Find the change in total charge on $\mathbb{R}$ and the change in charge on the fixed interval $[0,1]$. Does an identically conserved improvement leave every regional charge unchanged?

Solution. The divergence of the improvement is $\partial_\mu \partial_\nu K^{\mu\nu}$. Interchange μ and ν, commute the derivatives and use antisymmetry. The same sum equals its negative, so it is zero identically. The density change is $\partial_i K^{0i}$ because $K^{00} = 0$. On a periodic cube its opposite face values cancel. For compactly supported K^{0i} on $\mathbb{R}^d$, extend the spatial integral over a containing cube; every face value is zero.

For the stated function, direct differentiation gives

$$\widetilde{j}^0 - j^0 = -\frac{2tx}{(1+x^2)^2}, \qquad \widetilde{j}^1 - j^1 = -\frac{1}{1+x^2}.$$

Their divergence is zero, even though neither component vanishes. On $[-R,R]$ the integrated density is $K^{01}(t,R) - K^{01}(t,-R) = 0$. The density is integrable, so the total charge change is zero. On the finite interval the same primitive gives

$$\widetilde{Q}_{[0,1]} - Q_{[0,1]} = K^{01}(t,1) - K^{01}(t,0) = \frac{t}{2} - t = -\frac{t}{2}.$$

Thus regional charges can change. Their changed endpoint flux accounts for that change; identical local divergences do not remove endpoint terms.

Exercise 2.2 (The converse with an off-shell identity supplied). For a first-order density, suppose a legitimate characteristic Q^a and specified local current j^μ satisfy $\partial_\mu j^\mu = -E_a Q^a$ for arbitrary fields, before imposing their equations. Prove that $B^\mu = p_a^\mu Q^a - j^\mu$ gives $\delta_Q L = \partial_\mu B^\mu$.

Apply this calculation on $\mathbb{R}^{1,1}$ to $L = \frac{1}{2}(\phi_t^2 - \phi_x^2)$, $\eta(t,x) = t + x$, and

$$j^0 = \eta\phi_t - \phi, \qquad j^1 = -\eta\phi_x + \phi.$$

Find its characteristic, the divergence term and an actual one-parameter point transformation realising that characteristic. Verify the off-shell identity directly.

For context, the following broader correspondence is a *cited input*, not a task to prove: in the local finite-derivative formulation for the scalar wave Euler–Lagrange equation, conservation-law classes have generalized variational characteristics. Currents are identified modulo currents vanishing on solutions and identically conserved improvements; characteristics are identified modulo ones vanishing on solutions. Here local means depending smoothly on the coordinates, field and finitely many of its derivatives. This is the scalar wave case of Olver, §5.3, Theorem 5.58, p. 334. Explain why the assertion that an arbitrary current is conserved on solutions is not the supplied off-shell identity used in the first calculation.

Solution. Lecture 1 gives the identity $\delta_Q L = E_a Q^a + \partial_\mu(p_a^\mu Q^a)$. The supplied hypothesis substitutes $E_a Q^a = -\partial_\mu j^\mu$ into that formula. Hence

$$\delta_Q L = \partial_\mu(p_a^\mu Q^a - j^\mu), \qquad B^\mu = p_a^\mu Q^a - j^\mu.$$

This proves the stated algebraic converse with its off-shell hypothesis.

For the massless density, $p^0 = \phi_t$, $p^1 = -\phi_x$ and $E = -(\phi_{tt} - \phi_{xx})$. Since $\eta_t = \eta_x = 1$, differentiation of the given current yields

$$\partial_t(\eta\phi_t - \phi) + \partial_x(-\eta\phi_x + \phi) = \eta(\phi_{tt} - \phi_{xx}).$$

Thus its characteristic is $Q^1 = \eta$. The point transformation fixes spacetime and sends the field value u to $u + \varepsilon(t + x)$. It is smooth and invertible for every parameter. The recovered divergence terms are

$$B^0 = \phi_t\eta - j^0 = \phi, \qquad B^1 = -\phi_x\eta - j^1 = -\phi.$$

Direct variation also gives $\delta_Q L = \phi_t\eta_t - \phi_x\eta_x = \phi_t - \phi_x = \partial_t B^0 + \partial_x B^1$.

An on-shell conservation law only says that a divergence vanishes after substitution of a solution. It does not specify a characteristic Q^a or a local identity valid on arbitrary fields. The cited converse supplies a local representation under its stated class and equivalences. The elementary argument above starts after such a representation has been supplied; it is not a proof of that general theorem. Neither assertion alone ensures convergence or conservation of a full-space charge.

Exercise 2.3 (A Noether charge escapes to infinity). Let p and q be real fields on $\mathbb{R}_t \times \mathbb{R}_x$, with density $L = p(q_t + x^2 q_x)$. Choose $p_0 \in C_c^\infty((1,2))$, nonnegative and with integral one, and extend it by zero to $\mathbb{R}$. Set $q = 0$ and

$$p(t,x) = \begin{cases} (1+tx)^{-2} p_0\big(x/(1+tx)\big), & 1+tx > 0, \\ 0, & 1+tx \le 0. \end{cases}$$

(a) Derive the two field equations. Find the Noether current for the constant shift of q. Prove that the displayed fields are smooth at every finite spacetime point and satisfy both equations.

(b) Compute $Q(t) = \int_{\mathbb{R}} p(t,x)\mathrm{d}x$. Find its derivative for $t > 0$ and the limiting flux as $x \to +\infty$. Show that the initial charge is one but the charge is zero for $t \ge 1$.

(c) Identify exactly which full-space charge hypothesis in Lecture 2 fails. Why is this neither an off-shell example nor a claim about Klein–Gordon propagation?

Solution. The density has no derivatives of p, so $E_p = q_t + x^2 q_x$. Variation of q gives $E_q = -p_t - \partial_x(x^2 p)$. For $q \mapsto q + \varepsilon$, the characteristic components are $Q^q = 1$ and $Q^p = 0$, and the density is invariant. Its momentum components for q are $p_q^0 = p$ and $p_q^1 = x^2 p$. Therefore the Noether current is $j^0 = p$, $j^1 = x^2 p$.

On the open branch $1 + tx > 0$, put $\xi = x/(1+tx)$. Differentiating the displayed formula verifies $p_t + \partial_x(x^2 p) = 0$. Equivalently, the trajectories and density Jacobian satisfy

$$x(t) = \frac{\xi}{1 - t\xi}, \qquad x_t = x^2, \qquad \frac{\partial x}{\partial \xi} = (1 - t\xi)^{-2}, \qquad p(t, x(t)) = (1 - t\xi)^2 p_0(\xi).$$

On the other branch $p = 0$, so the equation holds there too. At a finite point with $1 + tx = 0$, necessarily $x \neq 0$. In a small neighbourhood of that point, whenever the denominator is small, $|x/(1+tx)| > 2$. The profile is then zero. Thus p is identically zero near the join and is smooth across it. At $t = 0$ the denominator is one and $p(0,x) = p_0(x)$. The field $q = 0$ solves its equation everywhere.

The support lies on $x > 0$. If $t \le 0$, the change of variables above reaches every $\xi > 0$. If $t > 0$, it reaches exactly $0 < \xi < 1/t$. The density times its Jacobian is $p_0(\xi)$, so

$$Q(t) = 1 \quad (t \le 0), \qquad Q(t) = \int_0^{1/t} p_0(\xi)\mathrm{d}\xi \quad (t > 0).$$

These integrals are finite. For $0 < t \le 1/2$ the value is one, and for $t \ge 1$ it is zero, because the initial support lies in $(1,2)$. FTC now gives $Q'(t) = -p_0(1/t)/t^2$ for $t > 0$. For $t > 0$, at the right spatial end,

$$\lim_{x \to +\infty} x^2 p(t,x) = \frac{p_0(1/t)}{t^2},$$

while the left flux is zero. This is exactly the lost charge rate with the outward sign of Lecture 2.

The nonnegative profile is positive somewhere. Hence for an open interval of times in $(1/2, 1)$ the right flux limit is strictly positive. On each such slice $x^2 p$ is not integrable over $\mathbb{R}$. The spacetime-integrable flux hypothesis of Lecture 2 therefore fails. The current is locally conserved on an actual smooth solution, and its slice density is integrable. The failure concerns its spatial infinity. This model has transport speed x^2 and is not the Klein–Gordon equation.